Decrease by a percentage

Task A: Decreasing an amount by a percentage

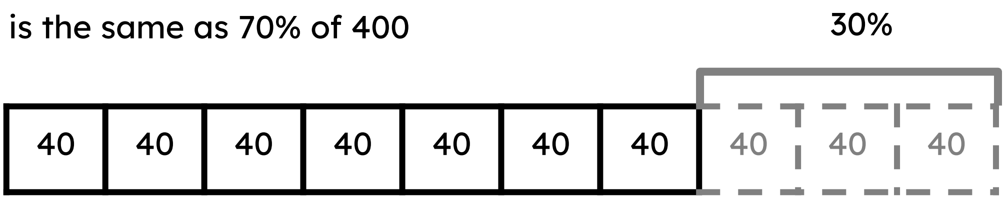
1) Pair up the equivalent questions:

Decrease 50 by 30%	Work out 80% of 50
Work out 97% of 50	Increase 50 by 45%
Decrease 50 by 20%	Work out 70% of 50
Work out 145% of 50	Decrease 50 by 3%

2) Fill in the bar model and missing information to show:

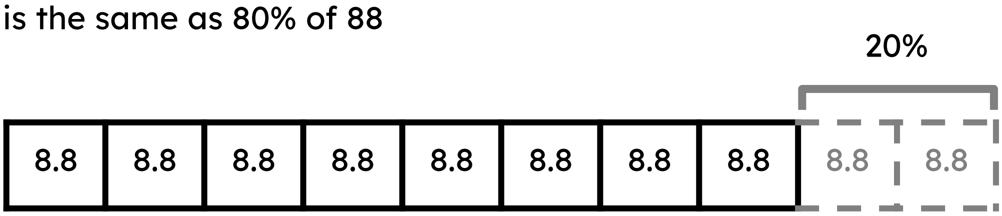
a) 400 decreased by 30% is the same as 70% of 400

Answer = *280*



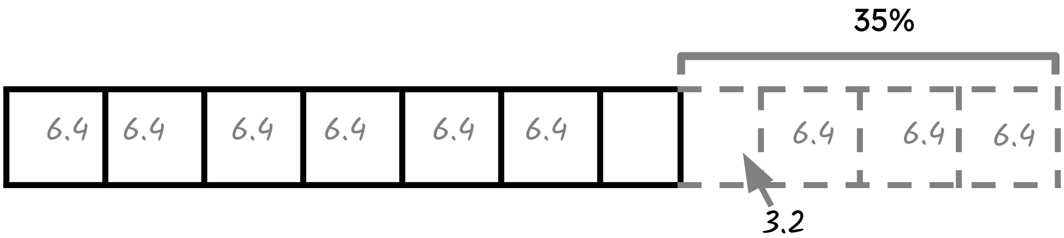
b) 88 decreased by 20% is the same as 80% of 88

Answer = *70.4*

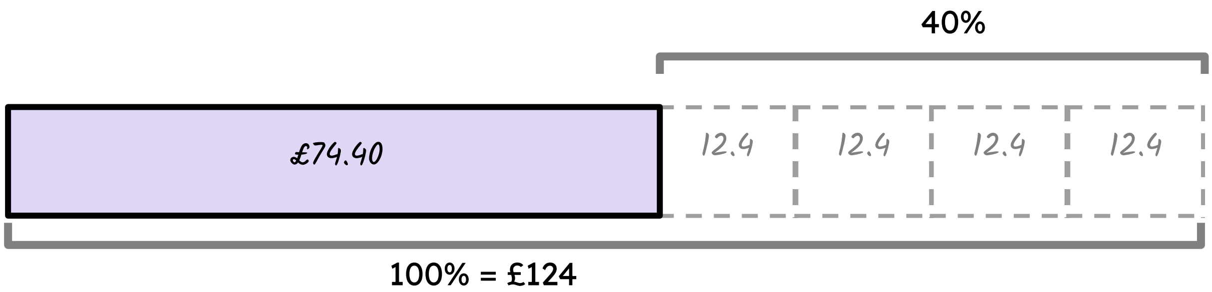


c) 64 decreased by 35% is the same as 65% of 64

Answer = *41.6*



3a) Draw a bar model to work out and represent £124 decreased by 40%



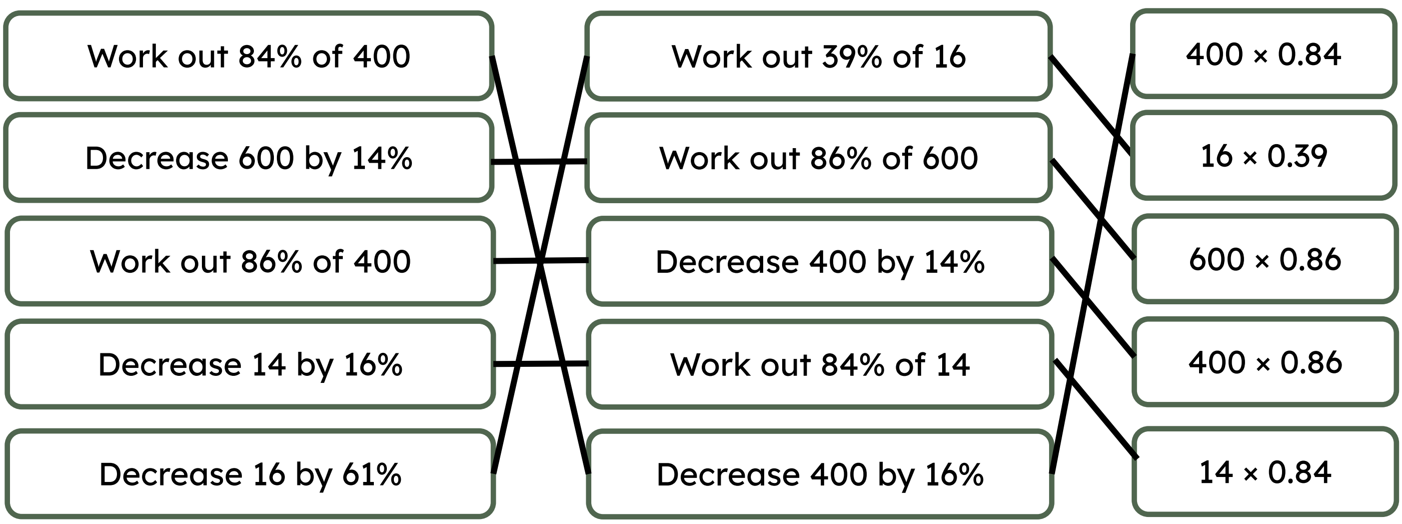
Decreasing £124 by 40%
= 124 − 12.4 × 4
= 74.4
£74.40

3b) Draw a bar model to work out and represent £72 decreased by 35%

Decreasing £72 by 35%
= 72 − 7.2 × 3 − 3.6
= 46.80
£46.80

Task B: Using a multiplier

1) Match the statements with the calculation.



2) Work out the amount of money when £94 has been decreased by 47%

$94 \times 0.53 = 49.82$

3) Work out the amount of money when £162 has been decreased by 28%

$162 \times 0.72 = 116.64$

4) A T-shirt was reduced by 24% in a sale.

How much is the T-shirt now?

£25 × 0.76 = £19

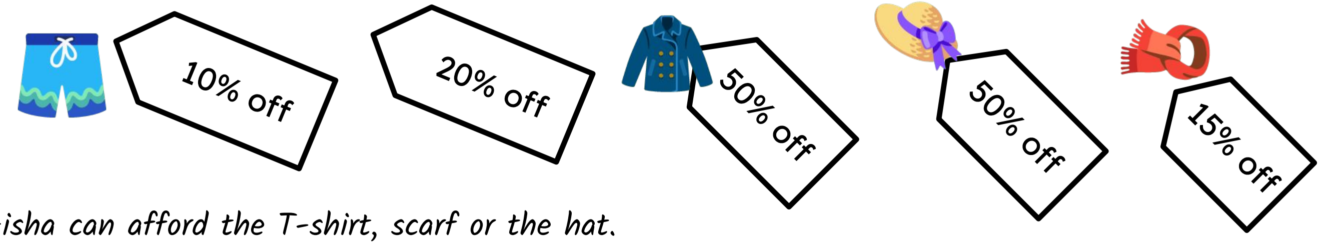
Task C: Mixed percentage change

1) Tick the calculations that correctly represent each statement.

Statement	Calculation	Tick
Increase £20 by 35%	20 × 1.35	✓
Decrease £20 by 35%	20 × 0.35	
Increase £35 by 20%	35 × 0.2	
Decrease £35 by 20%	35 × 0.8	✓
Increase £45 by 1%	45 × 1.1	
Decrease £45 by 1%	45 × 0.99	✓

2a) Aisha has £20 to buy a present for her dad. There is a sale on at the shop.
Below are some items with their original prices and the discounts to be applied.
Which of the items can she afford?

Shorts £30 T-shirt £25 Jacket £50 Hat £22 Scarf £22



Aisha can afford the T-shirt, scarf or the hat.

b) For those she can't afford, what discounts would she need so that she can afford any item in the shop?

Aisha would need these discounts: Shorts: a minimum of a 33.33% (2 d.p.) discount.
Jacket: a minimum of 60% discount.

3) Laura says,
"If you increase an amount by 13% and then decrease it by 13% you get the same value you started with."
Explain whether Laura is correct.

Laura is not correct. It can be explained in a few different ways: Increasing an amount by 13% then gives you a new, different amount. This new, different amount is then regarded as 100%, thus finding 13% of this new, different amount is not the same as 13% of the original amount.
Using multipliers, an amount increased by 13% then decreased by 13% is the same as × 1.13 × 0.87. Using the associative law, this is the same as × 0.9831 This means the final answer is 98.31% of the original amount.